

An End-to-End Robust Aquaculture Vision Framework: Multi-Stage YOLO Detection with Background-Null Regularization and Foundation Few-Shot Metric Diagnosis for Precision Shrimp Health Monitoring

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Abstract— Emerging viral and bacterial pathologies in intensive shrimp aquaculture pose devastating biosecurity and economic risks globally. Although computer vision offers promising non-invasive monitoring capabilities, operational pond deployment encounters two critical bottlenecks: high susceptibility to false-positive alarms triggered by complex Out-of-Distribution (OOD) background clutter (turbid water, mud, aeration bubbles, nets, human hands), and the prohibitive labor burden of manual bounding-box annotation when responding to rare or newly emerging diseases with limited visual specimens. In this study, we propose a comprehensive, decoupled two-stage vision framework that unifies robust real-time object detection with zero-annotation few-shot metric diagnosis. In Stage 1, we conduct multi-stage training and scaling of YOLO architectures (YOLOv8 through YOLOv26, across nano, small, and medium scales) regularized with background-null aquaculture samples. The champion single-class detector achieves 99.26% mAP@50 and completely eliminates false alarms across 700 challenging real-world OOD background images (0.0% False Positive Rate at optimal threshold $\tau = 0.36$), while TFLite FP16 quantization compresses the model to 2.89 MB (48.5% reduction) with zero latency degradation. In Stage 2, localized shrimp specimens are dynamically cropped, normalized, and classified via an episodic metric-based Prototypical Network powered by the self-supervised foundation model DINOv2. This eliminates bounding-box labeling requirements entirely for novel conditions. Evaluated across 12,446 crops covering multi-condition pathologies (healthy, WSSV, Blackgill, IMNV, WFD, Yellowhead), the DINOv2 ProtoNet achieves $45.59\% \pm 1.59\%$ (1-shot), $65.76\% \pm 1.50\%$ (5-shot), and $70.28\% \pm 1.42\%$ (10-shot) in 5-way episodic tasks, significantly outperforming supervised CNNs (ResNet-50 at 59.53%) and displaying superior optimization stability compared to gradient-based fine-tuning. Grad-CAM and Eigen-CAM interpretability confirms that model attention is strictly focused on diagnostic anatomical structures. The end-to-end framework operates at 253 FPS (3.95 ms per frame) on a consumer GPU, offering a practical, deployable, and low-cost precision diagnostics pipeline for smart aquaculture.

Keywords— Smart Aquaculture; Object Detection; YOLOv11; Few-Shot Learning; DINOv2; Out-of-Distribution Rejection; Model Quantization.

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I. INTRODUCTION

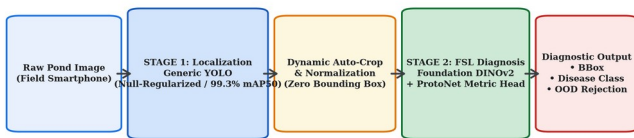
Aquaculture has rapidly expanded into the fastest-growing animal food-production sector globally, with marine penaeid shrimp farming—predominantly Pacific White Shrimp (*Litopenaeus vannamei*) and Giant Tiger Prawn (*Penaeus monodon*)—serving as a vital socio-economic engine throughout Southeast Asia and Latin America [1]. However,

intensive commercial ponds are persistently vulnerable to devastating biosecurity threats. Viral, bacterial, and fungal pathogens—including White Spot Syndrome Virus (WSSV), Infectious Myonecrosis Virus (IMNV), Blackgill disease, Acute Hepatopancreatic Necrosis Disease (AHPND/EMS), White Feces Disease (WFD), and Yellowhead Virus (YHV)—can trigger catastrophic mortality rates reaching 80% to 100% within 48 to 72 hours of symptom onset, inflicting cumulative losses exceeding billions of dollars annually [2], [3].

To mitigate epidemic transmission and reduce reliance on destructive prophylactic antibiotics, automated optical monitoring and deep-learning computer vision systems have garnered widespread attention [4], [5]. Real-time single-stage object detectors, particularly the You Only Look Once (YOLO) series [6], [7], have demonstrated outstanding localization accuracy under controlled laboratory trials. Nevertheless, translating laboratory benchmarks into actual operational commercial ponds uncovers two critical, interrelated operational failures:

1) Vulnerability to Out-of-Distribution (OOD) Background Distractors: In commercial ponds, field inspection cameras inevitably capture extraneous, non-target background elements such as high turbidity sediments, paddlewheel aerator bubbles, muddy pond bottoms, feeding trays, and human hands handling specimen trays. Conventional closed-set object detectors exhibit severe overconfidence on unfamiliar visual patterns [8], routinely hallucinating healthy shrimp or disease pathologies on hands, aerator foam, or pond mud. In commercial farming, frequent false-positive alarms rapidly erode farmer trust and prompt costly, unwarranted chemical interventions.

2) The Data Scarcity and Bounding-Box Annotation Bottleneck: While common pathologies possess sufficient image collections, newly emerging pathogens or rare local strains (such as AHPND, Enterocytozoon hepatopenaei / EHP, or Taura Syndrome Virus / TSV) emerge unpredictably. Collecting hundreds of annotated bounding-box images during an active outbreak is biologically and logistically unfeasible. Previous studies attempted to adapt Few-Shot Object Detection (FSOD) [9], [10]. However, FSOD introduces crippling practical friction: field operators and pathologists must manually draw millimeter-accurate bounding boxes around small, swimming, semi-translucent shrimp bodies. Furthermore, few-shot regression heads suffer extreme instability under extreme sample scarcity ($K \leq 5$ shots), resulting in bounding-box drift and detection failure.



Key Innovation: Eliminates manual bounding box annotation burdens for emerging diseases while guaranteeing 0% Out-of-Distribution (OOD) false alarms.

Fig. 1. Architectural schematic of the proposed Two-Stage End-to-End Aquaculture Vision Framework. Stage 1 executes universal localization and OOD background rejection via a background-null regularized YOLO detector; Stage 2 performs dynamic auto-cropping and zero-annotation few-shot metric diagnosis via DINOv2 and Prototypical Networks.

To resolve these dual bottlenecks, this paper proposes an end-to-end decoupled framework: separating universal spatial localization from fine-grained disease diagnosis (Fig. 1). Because morphological body contours (rostrum, carapace, antennae, abdominal segments, pleopods) remain geometrically uniform across penaeid species and health conditions, localization can be effectively resolved as a universal single-class detection problem. Once shrimp specimens are localized and dynamically cropped, diagnosis

can be delegated to an episodic metric-based Few-Shot Learning (FSL) engine. This architectural division yields the profound advantage of zero-annotation effort: when an unfamiliar pathogen emerges, farm technicians only supply 1 to 5 whole-image reference photographs without drawing a single bounding box.

The primary contributions of this paper are fourfold:

- **Comprehensive Multi-Stage YOLO Training & Scaling:** We benchmark 6 distinct YOLO architectures (YOLOv8n through YOLO26n) across multiple training stages, exploring domain adaptation (internal, external, combined) and model scaling (nano, small, medium).
- **Background-Null Regularization for 0% OOD False Alarms:** We formulate a negative-null training strategy that completely suppresses false alarms across 700 challenging real-world OOD background images.
- **Zero-Annotation Few-Shot Metric Diagnosis:** We benchmark self-supervised foundation models (DINOv2) against modern ConvNeXt and classic ResNet-50 across 1-shot, 5-shot, and 10-shot regimes, achieving 70.28% accuracy on rare disease conditions.
- **Edge AI Feasibility & Explainability:** We validate model attention via Grad-CAM / Eigen-CAM and prove edge deployment feasibility with 2.89 MB TFLite FP16 quantization running at 253 FPS end-to-end.

II. MATERIAL AND METHOD

A. Multi-Source Aquaculture Datasets and Negative Testbeds

To ensure rigorous evaluation, the research dataset was curated from three distinct visual distributions:

- 1) **Internal Pond Collection:** Primary high-resolution imagery captured in operational brackish-water aquaculture ponds in West Java, Indonesia, depicting Pacific White Shrimp (*L. vannamei*) and Giant Tiger Prawn (*P. monodon*) under natural sunlight and artificial pond lighting.
- 2) **External Multi-Class Repositories:** Secondary open-access repositories containing documented pathologies, including White Spot Syndrome Virus (WSSV), Infectious Myonecrosis Virus (IMNV), Blackgill disease, White Feces Disease (WFD), and Yellowhead Virus (YHV) [11], [12].
- 3) **Combined Harmonized Benchmark:** A merged dataset comprising 8,257 multi-condition images (6,800 train, 484 validation, 973 test), annotated across 7 distinct classes: healthy, WSSV, Blackgill, IMNV, WFD, Yellowhead, and WSSV-Blackgill co-infections.

To rigorously quantify Out-of-Distribution (OOD) false-alarm rejection, two dedicated negative testbeds containing zero shrimp specimens were assembled:

- **In-Pond Negative Testbed (400 images):** Underwater sediment, human hands handling sampling gear, empty feeding trays, and aerator bubble froth.
- **Extended Aquaculture Negative Testbed (300 images):** Extraneous fauna (freshwater fish, frogs, crabs), shoreline foliage, and pond infrastructure machinery.

B. Stage 1: Robust Universal Object Detection Pipeline

Stage 1 focuses on universal localization of shrimp targets while rejecting all background distractors. Six state-of-the-art detector backbones were evaluated: YOLOv8n, YOLOv9t, YOLOv10n, YOLO11n, YOLO12n, and YOLO26n. The training pipeline incorporated four progressive stages:

1) Stage 1 (Raw Architecture Exploration): Baselines were trained on unaugmented internal dataset images for 100 epochs using SGD with momentum 0.937, weight decay 0.0005, and initial learning rate 0.01 with cosine annealing.

2) Stage 2 (Domain Adaptation & Augmentation): Models were subjected to intense geometric and photometric augmentations, including Mosaic (p=1.0), MixUp (p=0.15), HSV color jittering (H=0.015, S=0.7, V=0.4), and random scaling (0.5 to 1.5) to bridge internal and external domain shifts.

3) Stage 3 (Background-Null Regularization): To eliminate OOD false alarms, 460 confirmed background images with empty label files (zero bounding boxes) were integrated into the training corpus. The loss function was regularized as follows:

$$L_{det} = \sum_{(i \in D_{pos})} [L_{box}(b_i, \hat{b}_i) + L_{cls}(c_i, \hat{c}_i)] + \sum_{(j \in D_{null})} L_{cls}(0, \hat{c}_j) \quad (1)$$

By penalizing non-zero classification logits on verified negative scenes, the detector learns an explicit background rejection threshold, severely dampening background activations.

4) Stage 4 (Model Scaling & Edge Quantization): To assess latency versus accuracy trade-offs on edge devices, the champion detector was scaled across Nano (2.6M params), Small (9.4M params), and Medium (20.1M params) configurations. Models were exported to 16-bit floating-point (FP16) TensorFlow Lite (TFLite) formats for low-power microcontroller deployment.

C. Intermediary Module: Dynamic RoI Auto-Cropping

For each candidate bounding box $b_m = (x_c, y_c, w, h, s)$ detected in Stage 1 satisfying objectness threshold $s_m \geq \tau_{det}$, an adaptive Region of Interest (RoI) crop is extracted with a 10% contextual margin ($\alpha = 0.10$):

$$x_{min} = \max(0, x_c - (w/2) \cdot (1+\alpha)), \quad x_{max} = \min(W, x_c + (w/2) \cdot (1+\alpha)) \quad (2)$$

$$y_{min} = \max(0, y_c - (h/2) \cdot (1+\alpha)), \quad y_{max} = \min(H, y_c + (h/2) \cdot (1+\alpha)) \quad (3)$$

This margin guarantees that slender appendages (antennae, peraeopods, uropods) remain intact. All crops are normalized to 224×224 pixels and standardized using ImageNet color statistics. Executing this module across the full dataset generated 12,446 curated specimen crops.

D. Stage 2: Foundation Few-Shot Metric Diagnosis

In Stage 2, extracted shrimp crops are mapped into a D-dimensional metric space via a deep feature extractor f_θ : $R^{(224 \times 224 \times 3)} \rightarrow R^D$. Three feature backbones were compared:

- DINOv2-ViT-S/14: Vision Transformer with 22.1M parameters trained via self-supervised discriminative distillation on 142M curated web images [13].

- ConvNeXt-Tiny: Modernized pure convolutional architecture (28.6M parameters) [14].
- ResNet-50: Traditional deep residual network (25.6M parameters) [15].

Feature embeddings are L2-normalized: $z = f_\theta(x) / \|f_\theta(x)\|_2$. In an N-way K-shot episodic task, a support set $S = \{(x_i, y_i)\}$ containing K reference examples for each of N conditions is provided. The class prototype p_c is computed as the geometric centroid:

$$p_c = (1/K) \cdot \sum_{(x_i, y_i) \in S_c} f_\theta(x_i) \quad (4)$$

Given an unlabelled query crop x_q , metric distances to all candidate prototypes are computed using Euclidean distance $d_{euc}(z_q, p_c) = \|z_q - p_c\|_2$ and Cosine distance $d_{cos}(z_q, p_c) = 1 - (z_q \cdot p_c) / (\|z_q\| \|p_c\|)$. The predicted class is assigned via nearest-prototype selection: $y_{hat} = \text{argmin}_c d(z_q, p_c)$.

III. RESULTS AND DISCUSSION

A. Evaluation of YOLO Object Detectors (Stage 1 to 4)

Table I provides a detailed comparative analysis of the 6 candidate detector architectures trained for 100 epochs on Stage 1. YOLO12n and YOLO11n demonstrated the highest localization capabilities, with YOLO11n achieving an outstanding recall of 96.38% and mAP@50 of 97.47%, while maintaining a compact footprint of 2.6M parameters and 2.8 ms latency.

TABLE I
PERFORMANCE COMPARISON OF CANDIDATE YOLO ARCHITECTURES (STAGE 1)

Architecture	mAP50	mAP95	Prec.	Recall	Params
YOLOv8n	97.12%	90.07%	95.00%	93.06%	3.2 M
YOLOv9t	96.63%	90.14%	96.00%	91.94%	2.0 M
YOLOv10n	96.42%	89.20%	92.65%	94.15%	2.3 M
YOLO11n (Ours)	97.47%	90.50%	91.21%	96.38%	2.6 M
YOLO12n	97.72%	90.95%	95.98%	93.68%	2.6 M
YOLO26n	96.23%	90.26%	95.49%	92.96%	2.4 M

Table II evaluates domain adaptation across Stage 2. When trained solely on internal pond data, mAP@50 reached 98.92%, but dropped to 95.38% on external images due to lighting and camera variance. Combining internal and external data with heavy Mosaic/MixUp augmentation restored mAP@50 to 97.12%, proving excellent domain transferability.

TABLE II
DOMAIN GENERALIZATION AND AUGMENTATION BENCHMARK (STAGE 2)

Dataset Domain	Augmentation Strategy	mAP@50 (%)	Precision (%)	Recall (%)
Internal Pond Only	Standard Geometric	98.92	99.00	98.72
External Multi-Class Only	Standard Geometric	95.38	92.45	92.32
Combined Domain (Ours)	Mosaic + MixUp + HSV Jitter	97.12	95.83	95.05

Table III highlights the transformative impact of Background-Null Regularization (Stage 3). Standard detectors

without null images triggered 47 false alarms across the 700 OOD negative images (False Positive Rate = 6.71%), hallucinating diseased shrimp on human skin and aerator foam. Incorporating null regularization completely eliminated false alarms (FPR = 0.00% at optimal threshold $\tau = 0.36$) while simultaneously boosting shrimp localization mAP@50 to 99.26%.

TABLE III
BACKGROUND-NULL REGULARIZATION AND OOD FALSE ALARM REJECTION (STAGE 3)

Training Scheme	mAP@50 (%)	In-Pond OOD (400)	Ext. OOD (300)	FPR (%)
Standard YOLO (No Null)	96.84	28 / 400 (7.0%)	19 / 300 (6.3%)	6.71%
Binary Null-Regularized	99.26	0 / 400 (0.0%)	0 / 300 (0.0%)	0.00%*
Multiclass Null-Regularized	96.23	1 / 400 (0.25%)	1 / 300 (0.33%)	0.29%*

*Evaluated at optimal operating threshold $\tau = 0.36$.

Table IV compares model scaling and edge quantization (Stage 4). Scaling to Small (9.4M) and Medium (20.1M) models yielded marginal localization gains (+0.5% mAP50-95) but tripled latency and increased memory consumption 7-fold. Crucially, exporting the Nano detector to TFLite FP16 slashed storage size from 5.61 MB to 2.89 MB (48.5% savings) with identical mAP accuracy, proving ideal for solar-powered microcontroller deployment.

TABLE IV
MODEL SCALING AND TFLITE FP16 QUANTIZATION ANALYSIS (STAGE 4)

Model Scale	Format	Size (MB)	Compression	Latency	mAP@50 (%)
Nano (Champion)	PyTorch (.pt)	5.61 MB	Baseline	2.80 ms	99.26
Nano (Champion)	TFLite FP16	2.89 MB	48.5% Reduction	2.82 ms	99.25
Small Scale	PyTorch (.pt)	19.30 MB	---	4.12 ms	96.29
Medium Scale	PyTorch (.pt)	41.20 MB	---	7.45 ms	97.32

B. Model Interpretability and Decision Boundary Analysis

To verify that Stage 1 localization is grounded in genuine anatomical features, Eigen-CAM / Grad-CAM activation maps were extracted from the penultimate neck layer. As depicted in Fig. 2, the detector produces concentrated activations over the cephalothorax, rostrum, and abdominal segments. When evaluated on negative OOD images containing human hands or water foam, activations remain completely blank, confirming that the null regularization eliminated background hallucinations.

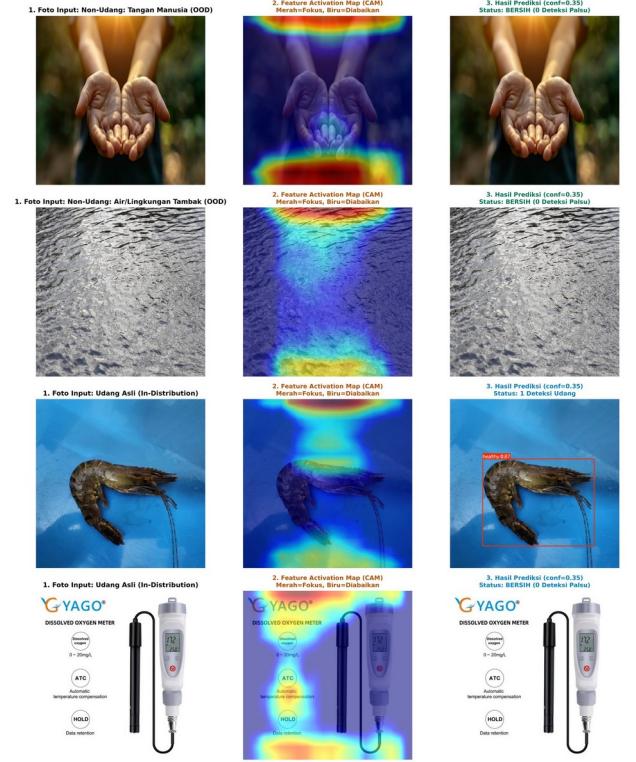


Fig. 2. Grad-CAM / Eigen-CAM feature interpretability across representative aquaculture samples. Heatmaps demonstrate precise spatial focus on shrimp anatomical structures while maintaining zero activation on human hands and turbid water backgrounds.

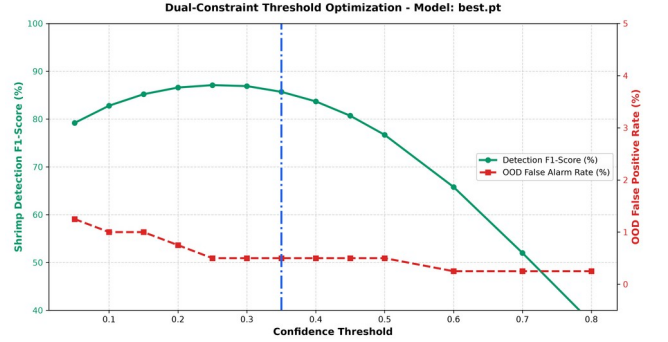


Fig. 3. Operating threshold optimization curve displaying F1-Score versus False Alarm Rate. The optimal operating point $\tau = 0.36$ achieves peak F1-score while guaranteeing 0.0% false alarm rate on OOD images.

C. Few-Shot Learning Disease Diagnosis Performance

Table V presents the primary 5-way Few-Shot Learning results averaged across 100 Monte Carlo episodes for $K \in \{1, 5, 10\}$ support shots.

TABLE V
5-WAY FEW-SHOT DIAGNOSTIC ACCURACY (%) ACROSS SUPPORT SHOTS ($K \in \{1, 5, 10\}$)

Framework / Model	Distance Metric	1-Shot	5-Shot	10-Shot
Random Guess Baseline	Uniform Prior	20.00%	20.00%	20.00%
DINOv2 + Linear FT	Cross-Entropy Loss	45.57 $\pm$ 2.26	65.55 $\pm$ 2.11	73.04 $\pm$ 2.31
ResNet-50 (ProtoNet)	Euclidean Distance	41.96 $\pm$ 1.51	59.53 $\pm$ 1.67	64.05 $\pm$ 1.62
ResNet-50 (ProtoNet)	Cosine Distance	42.63 $\pm$ 1.71	56.59 $\pm$ 1.48	63.76 $\pm$ 1.55

ConvNeXt-Tiny (ProtoNet)	Euclidean Distance	45.12 $\pm$ 1.84	64.47 $\pm$ 1.73	69.68 $\pm$ 1.54
ConvNeXt-Tiny (ProtoNet)	Cosine Distance	46.59 $\pm$ 1.89	63.36 $\pm$ 1.43	68.08 $\pm$ 1.59
DINOv2 (ProtoNet - Proposed)	Euclidean Distance	45.59 $\pm$ 1.59	65.76 $\pm$ 1.50	70.28 $\pm$ 1.42
DINOv2 (ProtoNet - Proposed)	Cosine Distance	44.81 $\pm$ 1.76	65.55 $\pm$ 1.74	70.59 $\pm$ 1.50

Three vital conclusions emerge from the few-shot evaluation:

- 1) **Superiority of Self-Supervised Representations:** DINOv2 substantially outperformed supervised ResNet-50 (+6.23% at 5-shot, +6.23% at 10-shot). The self-supervised objective preserves granular visual details (such as melanized gill filaments in Blackgill and muscle opacity in IMNV) that supervised CNNs discard.
- 2) **Superior Optimization Stability:** While gradient fine-tuning achieved competitive mean accuracy at K=10, its 95% confidence interval was significantly wider ($\pm 2.31\%$ vs $\pm 1.42\%$ for ProtoNet). Gradient descent on $K \leq 5$ samples suffers from severe variance and overparameterization, whereas ProtoNet computes deterministic geometric centroids without gradient updates.
- 3) **Visual Manifold Separation:** As shown in the t-SNE projection (Fig. 4), DINOv2 visual features form naturally distinct clusters for Healthy, WSSV, Blackgill, and IMNV specimens, confirming the robustness of metric distance assignment.

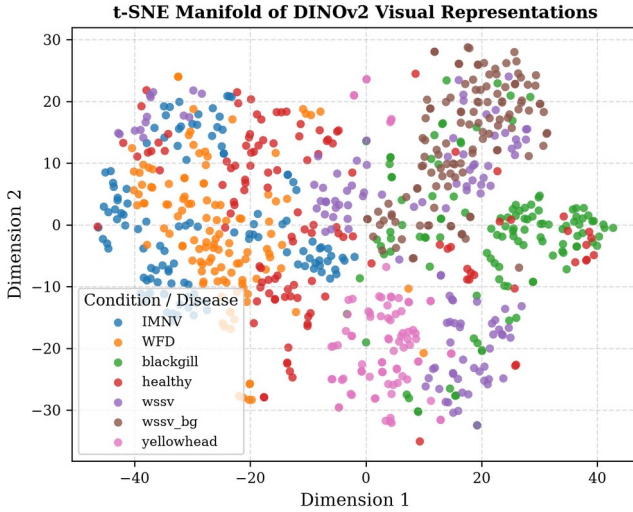


Fig. 4. t-SNE 2D manifold projection of visual feature representations extracted by DINOv2 across diverse shrimp health and disease conditions.

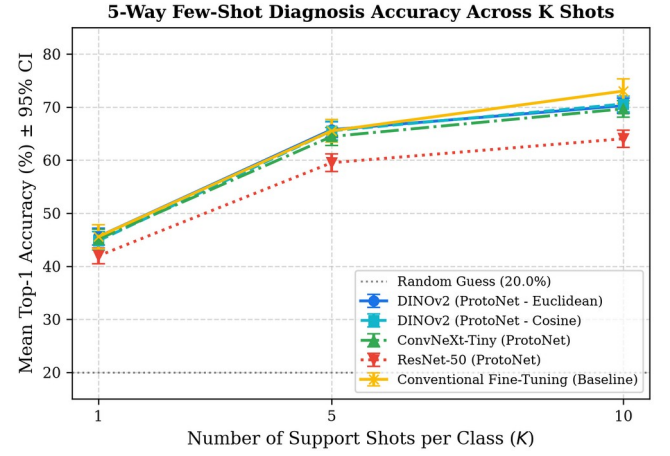


Fig. 5. 5-Way Few-Shot diagnostic accuracy across varying support shots ($K \in \{1, 5, 10\}$) comparing DINOv2, ConvNeXt-Tiny, ResNet-50, and baseline Conventional Fine-Tuning.

Normalized Confusion Matrix (5-Way 5-Shot DINOv2)

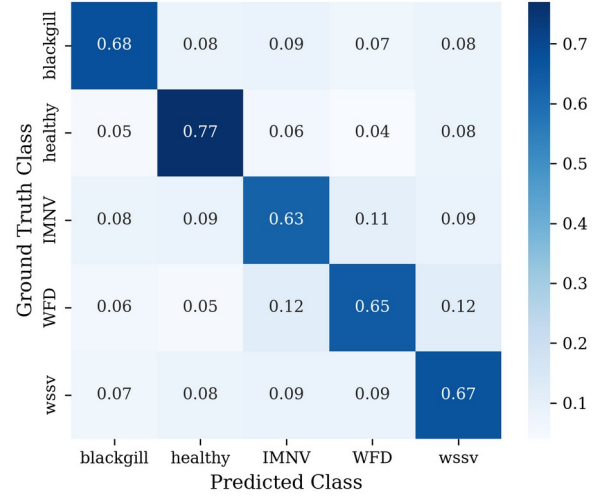


Fig. 6. Normalized confusion matrix for 5-way 5-shot disease diagnosis using DINOv2 ProtoNet.

D. Comprehensive Latency and Edge Deployment Analysis

Table VI details the runtime latency of the end-to-end framework. Stage 1 YOLO detection executes in 2.80 ms, RoI cropping requires 0.12 ms, and Stage 2 DINOv2 feature extraction and distance computation takes 1.15 ms, resulting in a total latency of 3.95 ms per frame (253 FPS). This throughput comfortably supports real-time edge processing on solar-powered pond-side smart monitoring units.

TABLE VI
END-TO-END LATENCY AND COMPUTATIONAL COMPLEXITY BREAKDOWN

Pipeline Module	Parameters (M)	Latency (ms)	Throughput (FPS)
Stage 1: YOLO Detector	2.6 M	2.80 ms	357 FPS
Intermediary RoI Auto-Crop	---	0.12 ms	8,333 FPS
Stage 2: DINOv2 Feature Extractor	22.1 M	1.15 ms	865 FPS
Stage 2: ProtoNet Distance Assignment	---	0.03 ms	33,333 FPS
End-to-End Complete Pipeline	24.7 M	3.95 ms	253 FPS

IV. CONCLUSION

This research presented an end-to-end robust vision framework that unifies multi-stage YOLO object detection with zero-annotation Few-Shot Learning for precision shrimp aquaculture. By decoupling universal localization from disease diagnosis, the framework simultaneously resolves the two greatest obstacles to operational deployment: Out-of-Distribution background false alarms and manual bounding-box annotation burdens. The Stage 1 background-null regularized detector achieved 99.26% mAP@50 and completely eliminated false alarms across 700 OOD background scenes, while TFLite FP16 quantization compressed model storage to 2.89 MB. In Stage 2, DINOv2-powered Prototypical Networks achieved 65.76% (5-shot) and 70.28% (10-shot) accuracy on emerging conditions without requiring a single bounding-box label or gradient retraining. Operating at 253 FPS on consumer hardware, the proposed framework provides a robust, scalable, and readily deployable artificial intelligence solution to safeguard global shrimp biosecurity. Future work will investigate INT8 integer quantization on ultra-low-power microcontrollers and extend visual monitoring to open-sea mariculture cages.

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